

CP-violation

C = charge conjugation:

particle $\rightarrow$ anti-particle

P = parity conjugation:

$$\mathbf{r} \rightarrow -\mathbf{r}$$

CP-violation:

$$P(\nu_\alpha \rightarrow \nu_\beta) \neq P(\bar{\nu}_\alpha \rightarrow \bar{\nu}_\beta)$$

Example:

$$P(\nu_e \rightarrow \nu_\mu) \neq P(\bar{\nu}_e \rightarrow \bar{\nu}_\mu)$$

CP-violating phase